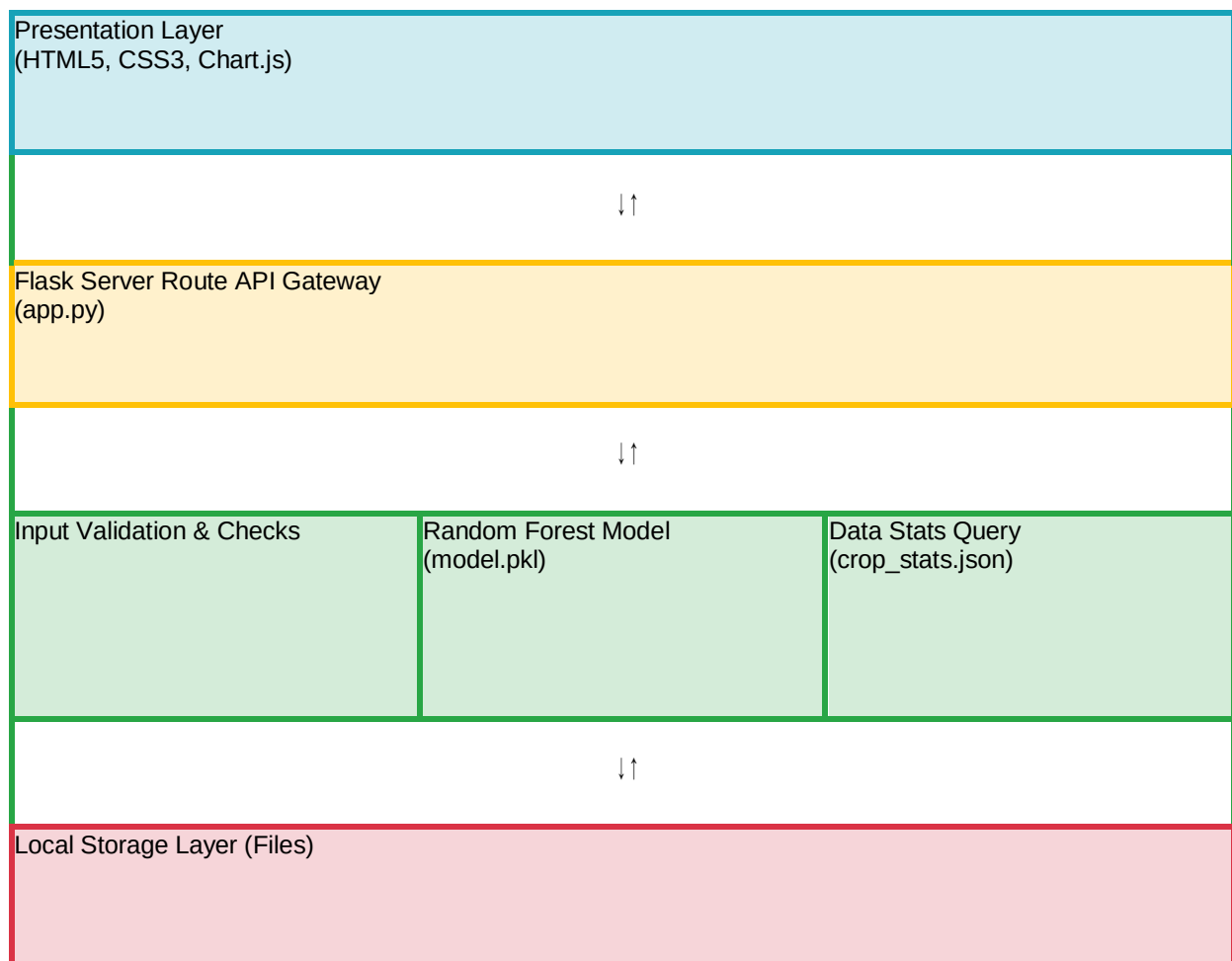


## Solution Architecture

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 2 July 2026                                                 |
| <b>Team ID</b>       |                                                             |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 5 Marks                                                     |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



## Component Description Table

| Component Name            | Description / Role in Architecture                                                     | Technologies Used                        |
|---------------------------|----------------------------------------------------------------------------------------|------------------------------------------|
| Presentation Layer        | Provides user interface, form controls, compatibility gauges, and Chart.js dashboards. | HTML5, CSS3, Bootstrap 5, Chart.js       |
| Application Routing / API | Handles HTTP requests, runs input validations, and exposes APIs for statistics data.   | Flask (Python 3.12)                      |
| Predictive Engine / Logic | Executes the Random Forest classifier model to output recommended crop name.           | Scikit-learn, joblib                     |
| Database & Configuration  | Caches crop summaries (mean/min/max) and raw dataset files.                            | crop_stats.json, Crop_recommendation.csv |